SEMINAR REFRESHER • AGES 10–17

BEYOND THE TRAINING SESSION

Athletic development isn't just about what you do during training; it's also about how you prepare and recover. This is a refresher on warm-up, cool-down, recovery and nutrition.

01 WARM-UP: USE THE RAMP PROTOCOL

A good warm-up gets your body and brain ready for the speed and movements of your sport. Build up gradually; around 10–15 minutes is a handy guide.

Whatever sport you play, a good warm-up follows the RAMP acronym. What changes is what you do in each step: it will look different for every sport.

R RAISE

Gradually lift your heart rate, breathing and body temperature.

e.g. jogging, skipping, side shuffles, backpedals

A ACTIVATE

Wake up the muscles you're about to use: hips, glutes, calves and core.

e.g. glute bridges, band walks, calf raises

M MOBILISE

Move your joints through the ranges basketball needs.

e.g. leg swings, hip and ankle mobility, lunge with a reach

P POTENTIATE

Build towards faster, more explosive, sport-specific movements.

e.g. low jumps building up, short sprints, sharp cuts, layups

SAME RAMP, DIFFERENT SPORT

Sprinter

Progressive runs and drills building up to near top speed.

Basketball player

Shuffles, jumps, landings, cuts and ball handling at rising intensity.

Weightlifter

Empty bar and lighter sets, building up to working weights.

A warm-up should prepare you for your sport, not just make you sweat.

02 COOL-DOWN, STRETCHING & MOBILITY

Three different tools that do three different jobs. Here's how to tell them apart.

AFTER TRAINING

WHAT IS A COOL-DOWN?

- A gradual shift from high-intensity exercise to lower-intensity movement.
- Light jogging, walking or gentle movement helps you settle.
- Match it to the session: hard game, a few easy minutes; light skills session, very little.

NO NEED FOR 15-20 MINUTES

The best thing you can do is cool down gradually: a light walk, some deep breathing or very light aerobic exercise. It may help keep the blood flowing and begin clearing the metabolic by-products built up during hard exercise.

STRETCHING: TWO DIFFERENT THINGS

DYNAMIC · IN YOUR WARM-UP

Active movements that prepare you for the actions your sport needs.

e.g. leg swings, walking lunges, high knees

STATIC · OVER TIME

Holding a position to improve flexibility over time. Hold gently and breathe; it should feel like a stretch, not sharp pain.

MOBILITY EXERCISES

Build the range ***and*** control needed for squatting, landing and changing direction.

e.g. deep squat holds, ankle rocks, split squats with a reach

Need help? Ask Murray

Choose exercises based on your own needs and the demands of your sport. A stiff-ankled player needs something different to a flexible player who needs better landing control.

ONE COMMON MYTH

THE MYTH

"STRETCHING PREVENTS MUSCLE SORENESS"

THE FACTS

Stretching can improve flexibility, but it has not been shown to consistently prevent or significantly reduce delayed-onset muscle soreness (DOMS), which peaks 1–3 days after new or harder exercise. For soreness, a gradual cool-down is more appropriate.

But that doesn't make stretching pointless. Flexibility is one piece of athletic performance: research on basketball players has linked better hamstring flexibility with higher vertical jumps. Stretching still has a valuable role in a broader athletic development program.

03 RECOVERY: FUNDAMENTALS FIRST

Two things matter most. The extras only come after.

SLEEP

8-10 hours a night for teens, on average

- Consistent sleep and wake times support recovery, learning and performance, and work with your natural body clock (circadian rhythm).
- Wind down before bed, cut late-night screens, and avoid late nights before early training.

Excessive screen time can delay sleep and reduce sleep quality, because your brain is still switched on and processing what's on the screen. More on this over the page.

TRAINING-LOAD MANAGEMENT



- Training is the stimulus; recovery is when your body adapts to it.
- You don't need to be in the best team, or in a million teams. Pick a couple of teams or programs where you belong and can keep growing, however good the team (or you) is. Development is what matters most.
- Playing several sports and programs at once (say football, domestic basketball, squad training and VJBL) leaves little time to recover and adapt. That can snowball into burnout and injury, and overuse injuries and stress fractures have been rising in youth sport.

THESE TWO ARE NON-NEGOTIABLE — No amount of foam rolling, massage, cold or warm water, compression garments or fancy tech will make up for poor sleep or poor training-load management.

THEN, THE EXTRAS

These may help you feel better, but none is essential.

Foam rolling
Cheap and an effective way to self-massage. Can help ease knots and general soreness.

Massage
Can enhance relaxation and recovery.

Cold-water immersion
Can help reduce soreness, especially during competition, like tournaments or hot days when you need to cool off fast and feel alert again. Most research is in adults, so it isn't essential for young athletes.

Warm-water immersion
Can promote relaxation and comfort, starting the recovery process after competition.

Compression garments
Little scientific evidence they improve performance or reduce soreness. Any benefit is mostly perceived: more or less placebo.

Don't spend money on recovery equipment while neglecting sleep, nutrition, and sensible training loads.

03 SCREENS, SLEEP & RECOVERY

Sleep is where recovery happens. What you do in the hour or two before bed sets up the whole night, so it's worth understanding what a late-night screen does to your brain and body.

LIGHT AND MELATONIN

Melatonin is the hormone that tells your body it's night. Bright evening light, including light from screens, can reduce melatonin and push your body clock later.

Teen body clocks already drift later, so late-night light can make it even harder to fall asleep.

A BUSY BRAIN

Games, social media, messages and videos keep your brain alert, reacting and often emotionally switched on.

Even after the screen goes off your brain needs time to settle, so you can end up lying there tired but wired.

LOST TIME

"Just one more video" pushes bedtime later, but school or early training still sets your wake-up time.

The result is shorter sleep, night after night, and less time for your body to recover and adapt.

Screen on late at night → Brain stays alert, melatonin is delayed → Fall asleep later, sleep is shorter or lighter → Less recovered at training next day

YOUR HORMONES AT BEDTIME

MELATONIN

Rises in the evening to help you feel sleepy. Evening light and late-night stimulation can delay it.

CORTISOL

A stress hormone that's meant to be low at night. Stress, stimulation and poor sleep can keep it higher, which makes it harder to wind down.

GROWTH HORMONE

Mostly released during deep sleep. It supports growth and repair, which is why good-quality sleep matters so much while you're growing.

WHAT THE RESEARCH SAYS

Most studies link more screen time, especially at night, with later bedtimes, shorter sleep and poorer sleep quality in young people. The size of the effect varies between studies and people.

In a study of 12–19 year olds, stopping screen use after 9 pm on school nights was linked with earlier sleep and more total sleep.

Among 112 middle and high school athletes, those sleeping under 8 hours a night were about 1.7 times more likely to have been injured. That shows a link, not proof of cause.

03 YOUR WIND-DOWN PLAN

You don't need to do everything at once. Pick two or three of these and build from there.

Screens off 30–60 minutes before bed

Aim for at least half an hour, and an hour or more is even better. Give your brain time to settle before you try to sleep.

Charge your phone outside the bedroom

No scrolling in bed. If you need an alarm, use a basic alarm clock so the phone stays out of reach.

Mute notifications at night

Use Do Not Disturb or sleep mode so messages don't wake you or tempt you to check.

Dim the lights in the evening

Lower lighting in the last hour or two tells your body night is coming. Bright daylight in the morning helps set your body clock.

Build a wind-down routine

A warm shower, some reading, light stretching or slow deep breathing, and packing your bag for tomorrow all work well.

Keep a consistent wake-up time

Try to wake at about the same time every day, including weekends. Consistent sleep and wake times support recovery, learning and performance.

Night mode helps a little, but it isn't a fix. Blue-light filters may reduce some of the light effect, but the content can still keep your brain switched on. Turning screens off works best.

Waking in the night? Resist checking your phone. The light and content can wake your brain up, and make it harder to drift back off.

FOR PARENTS

MAKE IT A FAMILY HABIT

Agree a charging spot outside bedrooms, set a screens-off time that applies to everyone, and model it yourself. Teens are far more likely to stick with it when the whole house does. If your teen has trouble sleeping for weeks, speak with your GP.

THIS WEEK'S CHALLENGE

ONE WEEK, SCREENS OFF EARLY

Try switching screens off 45 minutes before bed for a week. Notice how you fall asleep and how you feel at training the next day.

04 NUTRITION: FUEL YOUR PERFORMANCE

Eat a variety of foods, eat regularly, and fuel your training. You don't need an expensive diet or supplements to become a better athlete.

CARBOHYDRATES
Energy

Feeling tired and flat? It can come from not eating enough carbohydrate. You're growing, heading into or going through puberty and playing lots of sport, so your energy needs are already high. With the extra training and competition on top, you need to increase your energy supply to grow, adapt and keep improving.

e.g. bread, rice, pasta, oats, fruit

PROTEIN
Growth and repair

Your muscles are made of protein, so you need it to grow and repair the muscles you've used in training and games.

e.g. eggs, chicken, fish, yoghurt, beans

HEALTHY FATS
Growth and normal body functions

Healthy fats support joint health, heart and cardiovascular health, and brain health.

e.g. nuts, avocado, seeds, olive oil

HYDRATION
Supports performance

Dehydration can lead to more fatigue, poorer decision making and reduced force output, because the signals from your brain to your muscles don't work as well.

Drink water regularly all day, bring a bottle to training and games, and notice thirst. Needs vary from person to person.

Sports drinks are generally not necessary; water is the go-to. They can be useful to replenish carbohydrates quickly across a weekend with two or three games in a day, or when exercise demand is very high and carbohydrate stores are running low.

SIMPLE IDEAS

- **Before a big tournament, training day or camp:** eat a larger amount of carbohydrate than usual.
- **Before training:** something with carbohydrate you tolerate well, like a banana or toast.
- **After training:** carbohydrate and protein, like yoghurt and fruit or a chicken sandwich.
- Don't skip breakfast.

KEEP IT SIMPLE

- Eat a variety of foods and eat regularly through the day.
- Growing athletes need enough food for growth and training. Skipping meals or "dieting" is not a good idea.
- You don't need expensive diets or supplements to become a better athlete.
- Worried about growth, weight or eating habits? Chat with your GP or an Accredited Practising Dietitian.

05 KEY TAKEAWAYS

1
PREPARE PROPERLY
Warm up with purpose using RAMP, shaped to your sport.

2
RECOVER INTELLIGENTLY
Prioritise sleep, manage training loads, switch screens off early, and use extra strategies when appropriate.

3
FUEL PERFORMANCE
Eat a varied diet, eat enough to support growth and training, and stay hydrated.

SMALL HABITS, REPEATED CONSISTENTLY, CREATE BETTER ATHLETES.

A SIMPLE DAY, START TO FINISH

BEFORE TRAINING
- RAMP warm-up
- Carbohydrate snack you tolerate well
- Water bottle packed

DURING
- Sip water regularly
- Build up gradually, then go

AFTER
- Gradual cool-down: light walk or deep breaths
- Meal or snack with carbohydrate and protein

TONIGHT
- Screens off 30–60 minutes before bed
- Wind down
- Aim for 8–10 hours

This week's challenge: pick **one** habit and stick with it: a proper RAMP warm-up before every session, a consistent bedtime with screens off early, or always bringing a water bottle.

NEED HELP? ASK MURRAY

Not sure which warm-up, mobility work or recovery habits suit you? Book a call with Murray at Myers Sports Coaching.